# Infusion Information Sheet

***Please note this does not replace the Parenteral manual***

| Drug | To Mix Infusion | Conc/Volume | Run in: | Usual recommended dosage | Comments |
|---|---|---|---|---|---|
| **Amiodarone**<br><br>**CARDIAC ARREST DOSE:**<br>**300mg IV push (no dilution)** | **1. Loading dose:** Add 150mg(3ml) to 100ml D5W.<br><br>**2.** Add 900mg (18ml of 50mg/ml) to 500ml D5W in polyolefin bag **(Step 1 and 2)** | **1.** 1.5mg/ml<br><br>**2.** 1.8 mg/ml | ml/hr<br><br>mg/min | **1.Loading dose**: 150mg in 100ml D5W over 10min<br><br>**2. Step 1**. Then infuse at 1mg/min x 6 hours<br><br>**Step 2.** Then infuse at 0.5mg/min x18hrs | For ventricular arrhythmias. **Use filtered tubing** |
| **Dobutamine** | Withdraw 40ml from 500ml bag D5W or NS and discard. Add 500mg (40ml of 12.5mg/ml) to bag. | 1mg/ml | mcg/kg/min | Infusion: 2-20mcg/kg/min | |
| **Dopamine** | Premixed 400mg in 250ml D5W | 1600mcg/ml | mcg/kg/min | 1-5mcg/kg/min IV initially, with gradual increases of 1-4mcg/kg/min at 10-30min intervals **UP TO** 20-50mcg/kg/min | |
| **Epinephrine** | **1. Push dose:** Add 1mg (1mg/ml) to 100ml NS<br><br>**2.** Add 1mg (1ml of 1mg/ml) to 250ml D5W or NS | **1.** 10mcg/ml<br><br>**2.** 4mcg/ml | mcg/ml<br><br>mcg/min | **1.** 10-20mcg/dose. Withdraw from bag and use as a push dose.<br><br>**2.** Infusion: 2- 10mcg/min | |
| **Esmolol** | Premixed 2,500mg in 250ml.<br>Esmolol vials for initial load: 100mg/10ml | 10mg/ml | mg/kg/ min | **Initial load** 0.5mg/kg IV over 1min.<br>**Maintenance infusion** 0.05mg/kg/min for 4 min. If no response repeat load of 0.5mg/kg and increase infusion to 0.1/mg/kg/min | **MAX** infusion rate 0.3mg/kg/ min |
| **Fentanyl** | Premixed fentanyl 5mg in 100 mL | 50mcg/ml | mcg/hr | Infusion: 25-200mcg/hr | |
| **Flumazenil** | Withdraw 10ml from 50ml D5W or NS and discard, add 1mg (10ml from 0.5mg/5ml vials) to bag | 0.02mg/ml | mg/hr | Infusion: 0.1-0.4mg/hr | See parenteral manual for initial push dose |
| **Hydromorphone** | Add 20mg (2ml of 10mg/ml) to 100ml D5W or NS | 200mcg/ml (0.2mg/ml) | mcg/hr | Initiate with 200mcg/hr and titrate.<br>Usual range: 500-3000mcg/hr (0.5-3mg/hr) | |
| **Insulin (Humulin R or Novolin Toronto)** | Add 50 units (0.5ml of 100u/ml) to 50ml NS | 1 unit/ml | units/hr | Usual starting dose 0.1units/kg/hour (5-10 units/h) - Varies per patient, see physician orders | Needs dual sign-off |
| **Isoproterenol** | Add 1mg to 250ml of D5W or NS | 4 mcg/ml | mcg/min | **Initial infusion rate:** 5mcg/min (or as per physician orders) for bradycardia | |
| **Ketamine** | Withdraw 8ml from 100 ml D5W or NS and discard. Add 400mg (8ml of 100mg/2ml) to the bag. | 4mg/ml | mg/hr | **Analgesia/sedation continuous infusion:** 0.05-1.2mg/kg/hr.<br>*Needs to be calculated with patient weight to obtain mg/hr on pump. | See manual for more dosage indications/ load dose. |
| **Labetalol** | Withdraw 10ml from 250ml bag of D5W or NS and discard. Add 300mg (60ml from 5mg/ml) to the bag. | 1mg/ml | mg/min | Initiate rate at 0.5-2 mg/min<br>Adjust according to BP, up to 8mg/min has been used.<br>**Select "**all other units**" on pump<br><br>**MAX** cumulative dose of 300mg. | See manual for fluid restricted pt Pt should be supine. |
| **Lidocaine**<br><br>**CARDIAC ARREST LOAD:**<br>**1-1.5mg/kg** | Premixed bag of 1g in 250ml D5W. | 4mg/ml | mg/min | Continuous infusion: 1-4mg/min<br><br>**MAX total loading dose:** 3mg/kg | **Antidote:** Intralipids |
| **Magnesium**<br><br>**CARDIAC ARREST ONLY:**<br>**1-2g in 10ml D5W or NS. Administer over 1-2 min.** | Mix with compatible solution of NS or D5W.<br><br>(I.e: 2g in 50ml) | Max concentration 200mg/ml | g/hr | Max rate of 1g/hr (for stable patients).<br><br>Infusions can range from 15-60 min in different circumstances, follow physician order.<br><br>Caution: Hypotension | See parenteral manual for different dosage indications |

# Infusion Information Sheet

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|---|---|---|---|---|---|
| **Midazolam** | Withdraw 20ml from 100ml D5W or NS bag and discard. Then add 100mg (20ml of 5mg/ml) to bag | 1mg/ml | mg/hr | Infusion: 1-20mg/hr<br><br>Upper hard limit of 30mg/hr | |
| **Milrinone** | Withdraw 20 ml from 100ml D5W or NS bag and discard. Then add 20 mg (20 ml from 1 mg/ml) to the bag | 200mcg/ml | mcg/kg<br><br>mcg/kg/ min | **Optional loading dose**: 50mcg/kg over 10 min.<br><br>**Infusion** 0.375-0.75mcg/kg/min | **** Titration to be done ONLY with a physician's order**** |
| **Naloxone** | Withdraw 10ml from 100ml bag D5W or NS and discard. Add 12mg (30ml of 0.4mg/ml, 10ml vials to 90ml bag). | 0.1mg/ml | mg/hr | *Select **"high dose"** on pump<br><br>For high dose infusion, it is recommended to start rate at 1mg/hr. Titrate to patient's response. | |
| **Nitroglycerin** | Use a premixed bottle of 50mg in 250ml. | 200mcg/ml | mcg/min | **Initiate with:** 5-10mcg/min<br>Increase q3-5min by 5mcg/min.<br><br>Start high (up to 200mcg/min as per physician order) for hypertensive CHF. | **Use NON-PVC tubing + vented tubing** |
| **Norepinephrine** | Add 8mg (8ml of 1mg/ml) to 250ml NS | 32mcg/ml | mcg/min | Infusion: 8-12 mcg/min<br>Titrate for clinical response<br>Usual maintenance dose: 2-4mcg/min | *Use large vein - Extravasation Hazard* |
| **Phenylephrine** | **1. Push Dose:** Add 10 mg (1 ml from 10 mg/mL vial) to 100 mL of D5W or NS<br><br>**2.** Add 100 mg to 250 mL of NS | **1.** 100mcg/ml<br><br>**2.** 400mcg/ml | mcg/ml<br><br>mcg/min | **1.** 100mcg/ml dose. Withdraw from bag and use as a push dose.<br><br>**2.** Recommendation to start infusion at 100-180mcg/min | |
| **Procainamide** | **For loading dose only.** Withdraw 10ml from 500cc NS or D5W. Add 1000mg (10ml of 100mg/ml) to bag. | 2mg/ml<br><br>(Max concentration 20mg/ml) | mg/min | **Loading dose rate:** 20-50 mg/min<br><br>**STOP** infusion if arrhythmia converts, widening QRS (>50%) or hypotension.<br>If hypotension get order for bolus | **See RAFF-4 study for different conc/mixing guidelines** |
| **Propofol** | Ready-to-use vial 1000mg/100ml | 10mg/ml | mg/kg/ hr | Infusion: 0.3- 3mg/kg/hr - titrate to meet sedation goals<br><br>***Reminder to obtain orders for analgesia also.** | Use vented, non-filtered tubing |
| **Tranexamic acid (TXA)** | **1.** Add 1000mg (1g) to 100mL NS<br><br>**2.** Add 1000mg to 500mL D5W or NS | **1.** 10mg/ml<br><br>**2.** 2mg/ml | ml/hr<br><br>ml/hr | **1.** Loading dose 1000mg over 10mins<br><br>**2.** 62.5ml/hr x 8 hours (select intermittent infusion on pump) | **DO NOT PUSH** |
| **Vasopressin** | Withdraw 50ml from 250mL bag D5Wor NS and discard. Add 20 units (1ml of 20units/ml) to bag | 0.1unit/ml | unit/min | Infusion: 0.01-0.1units/min | See parenteral for different dosage indications |

## Y – Site Compatiblity Chart for Critical Care Drugs

| | DOBUTAMINE | DOPAMINE | EPINEPHRINE | FENTANYL | HYDROMORPHONE | MIDAZOLAM | NOREPINEPHRINE | PHENYLEPHRINE | PROPROFOL | VASOPRESSIN |
|---|---|---|---|---|---|---|---|---|---|---|
| DOBUTAMINE | | C | C | C | C | X | C | C | C | C |
| DOPAMINE | C | | C | C | C | C | C | C | C | C |
| EPINEPHRINE | C | C | | C | C | C | C | C | C | C |
| FENTANYL | C | C | C | | C | C | C | C? | C | C? |
| HYDROMORPHONE | C | C | C | C | | C | C | C | C | C |
| MIDAZOLAM | X | C | C | C | C | | C | C | X | C |
| NOREPINEPHRINE | C | C | C | C | C | C | | C | C | C |
| PHENYLEPHRINE | C | C | C | C? | C | C | C | | C? | C |
| PROPOFOL | C | C | C | C | C | X | C | C? | | ND |
| VASOPRESSIN | C | C | C | C? | C | C | C | C | ND | |

**LEGEND:**

**C:** Compatible using TOH standard drug concentrations.

**C?:** Compatible but using lower drug concentrations than standard TOH drug concentrations; unknown if compatible at TOH standard drug concentrations; do not mix.

**X:** Incompatible or conflicting information using TOH standard drug concentrations; do not mix.

**ND:** No data; do not mix.

**** Please see Lexicomp for full compatability through EPIC**